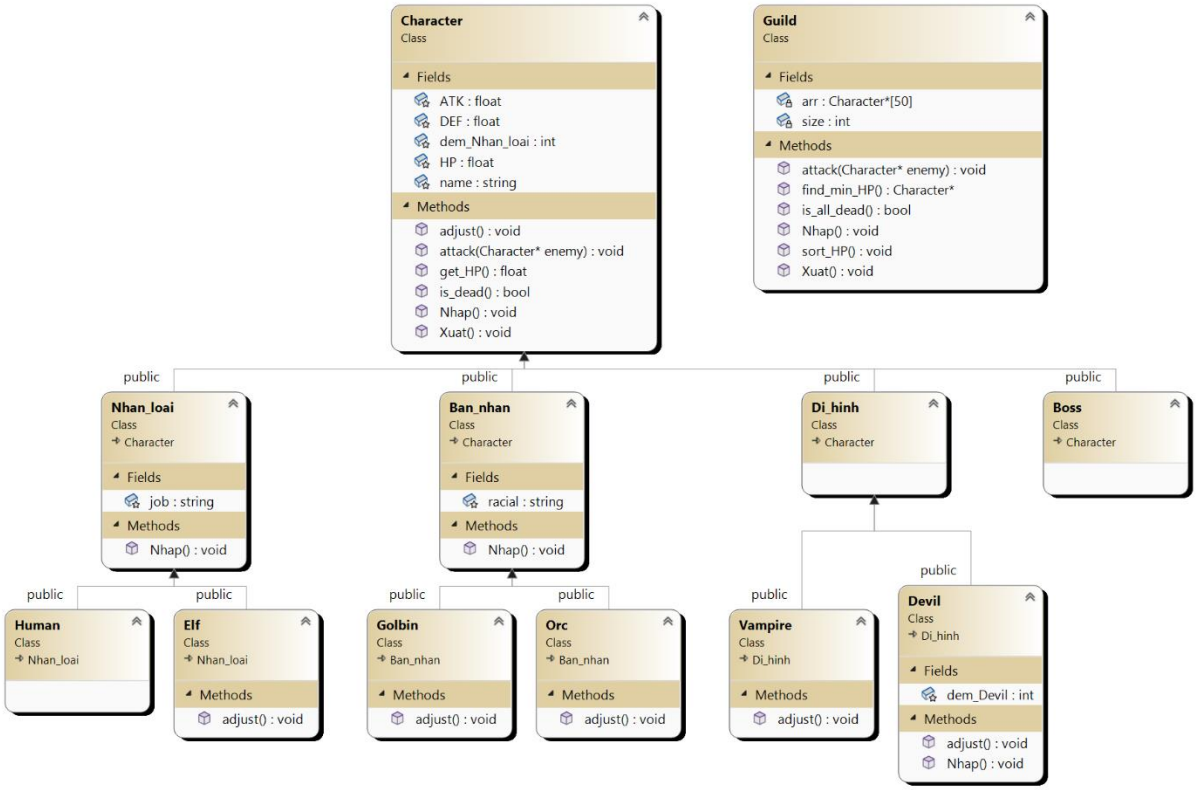
Bài 2.

Sơ đồ lớp:



Chương trình:

```
#include <iostream>
#include <string>
using namespace std;

class Character {
protected:
    static int dem_Nhan_loai;
    string name = "Unknown";
    float HP = 0.0;
    float ATK = 0.0;
    float DEF = 0.0;

public:
    float get_HP() { return HP; }
    bool is_dead() { return HP <= 0; }
    void attack(Character* enemy) {
        if (ATK > enemy->DEF) {
            enemy->HP = enemy->HP - (ATK - enemy->DEF);
        }
    }
```

```
    virtual void adjust() {
        return;
    }
    virtual void Nhap() {
        cin >> name >> HP >> ATK >> DEF;
    }
    void Xuat() {
        cout << name << " " << HP << endl;
    }
};

int Character::dem_Nhan_loai = 0;

class Boss : public Character {};

class Nhan_loai : public Character {
protected:
    string job = "Unknown";
public:
    void Nhap() {
        Character::Nhap();
        cin >> job;
        ++dem_Nhan_loai;
    }
};

class Ban_nhan : public Character {
protected:
    string racial = "Unknown";
public:
    void Nhap() {
        Character::Nhap();
        cin >> racial;
    }
};

class Di_hinh : public Character {};

class Human : public Nhan_loai {};

class Elf : public Nhan_loai {
public:
    void adjust() {
        DEF = DEF * 1.5;
    }
};

class Orc : public Ban_nhan {
public:
    void adjust() {
```

```
        HP = HP * 1.5;
    }
};

class Golbin : public Ban_nhan {
public:
    void adjust() {
        ATK = ATK * 1.1;
    }
};

class Vampire : public Di_hinh {
public:
    void adjust() {
        HP = HP + HP * 0.1 * dem_Nhan_loai;
    }
};

class Devil : public Di_hinh {
protected:
    static int dem_Devil;
public:
    void Nhap() {
        Character::Nhap();
        ++dem_Devil;
    }
    void adjust() {
        ATK = ATK + ATK * 0.1 * dem_Devil;
        DEF = DEF + DEF * 0.1 * dem_Devil;
    }
};

int Devil::dem_Devil = 0;

Character* init(int type) {
    switch (type) {
    case 1:
        return new Human;
    case 2:
        return new Elf;
    case 3:
        return new Orc;
    case 4:
        return new Golbin;
    case 5:
        return new Vampire;
    case 6:
        return new Devil;
    }
    return new Character;
```

```
}

class Guild {
private:
    int size = 0;
    Character* arr[50];
public:
    void Nhap() {
        cin >> size;
        for (int i = 0; i < size; ++i) {
            int loai;
            cin >> loai;
            arr[i] = init(loai);
            arr[i]->Nhap();
        }
        for (int i = 0; i < size; ++i) {
            arr[i]->adjust();
        }
    }
    void Xuat() {
        for (int i = 0; i < size; ++i) {
            if (!arr[i]->is_dead()) {
                arr[i]->Xuat();
            }
        }
    }
    void attack(Character* enemy) {
        for (int i = 0; i < size; ++i) {
            if (!arr[i]->is_dead()) {
                arr[i]->attack(enemy);
            }
        }
    }
    bool is_all_dead() {
        for (int i = 0; i < size; ++i) {
            if (!arr[i]->is_dead()) {
                return false;
            }
        }
        return true;
    }
    Character* find_min_HP() {
        int min_index = 0;
        for (int i = 0; i < size; ++i) {
            if (!arr[i]->is_dead()) {
                min_index = i;
                break;
            }
        }
        for (int i = 0; i < size; ++i) {
```

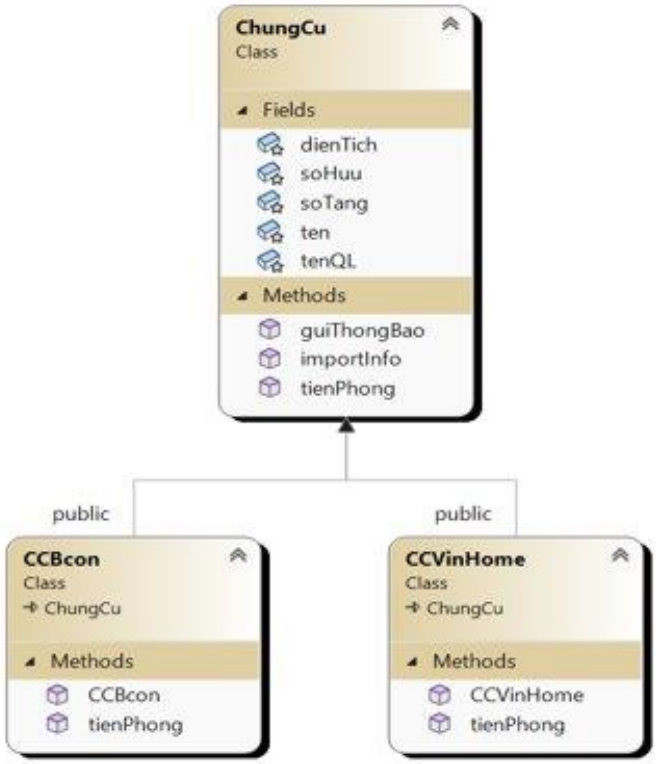
```
            if (!arr[i]->is_dead() && arr[i]->get_HP() < arr[min_index]->get_HP()) {
                min_index = i;
            }
        }
        return arr[min_index];
    }
    void sort_HP() {
        for (int i = 0; i <= size - 2; ++i)
            for (int j = i + 1; j <= size - 1; ++j)
                if (arr[j]->get_HP() < arr[i]->get_HP())
                    swap(arr[i], arr[j]);
    }
};

int main() {
    Guild team;
    team.Nhap();
    Boss boss;
    boss.Nhap();
    do {
        team.attack(&boss);
        if (boss.is_dead()) {
            break;
        }
        boss.attack(team.find_min_HP());
    } while (!team.is_all_dead());
    if (!boss.is_dead()) {
        boss.Xuat();
    }
    else {
        team.sort_HP();
        team.Xuat();
    }
    return 0;
}

/* Test case:
3
1 Adam 100 10 10 NongDan
2 Vi 200 10 10 CungThu
6 Jacob 400 50 30
Ainz 200 50 20
*/
```

Bài 3.

Sơ đồ lớp:



Code:

```
#include <iostream>
#include <string>
#include <time.h>
using namespace std;

int random(int min, int max);

class ChungCu {
protected:
    string ten;
    int soTang;
    double dienTich;
    string tenQL;
    string soHuu;
public:
    void importInfo();
    string guiThongBao();
    virtual double tienPhong() = 0;
};

class CCVinHome : public ChungCu {
public:
    CCVinHome();
    double tienPhong();
};

class CCBcon : public ChungCu {
```

```
public:
    CCBcon();
    double tienPhong();
};

void ChungCu_import(ChungCu**, int, int);
void ChungCu_listExport(ChungCu**, int);
void ChungCu_tienPhong(ChungCu**, int);

int main() {
    int VinHome_Amount, Bcon_Amount;

    cout << "So chung cu VinHome la: ";
    cin >> VinHome_Amount;
    cout << "So chung cu Bcon la: ";
    cin >> Bcon_Amount;
    system("cls");

    int size = VinHome_Amount + Bcon_Amount;
    ChungCu** a = new ChungCu * [size];

    ChungCu_import(a, VinHome_Amount, Bcon_Amount);
    ChungCu_listExport(a, size);
    ChungCu_tienPhong(a, size);
    system("pause");
}

int random(int min, int max) {
    static bool first = true;
    if (first) {
        srand(time(NULL));
        first = false;
    }
    return min + rand() % ((max + 1) - min);
}

void ChungCu::importInfo() {
    cout << "  Ten chung cu: ";
    getline(cin, ten);
    cout << "  So tang: ";
    cin >> soTang;
    cout << "  Dien tich chung: ";
    cin >> dienTich;
    cin.ignore();
    cout << "  Ten quan ly: ";
    getline(cin, tenQL);

}
string ChungCu::guiThongBao() {
    return tenQL + " - " + ten + " - " + soHuu;
}

CCVinHome::CCVinHome() {
    soHuu = "Vinhomes";
}
double CCVinHome::tienPhong() {
    if (dienTich > 600)
```

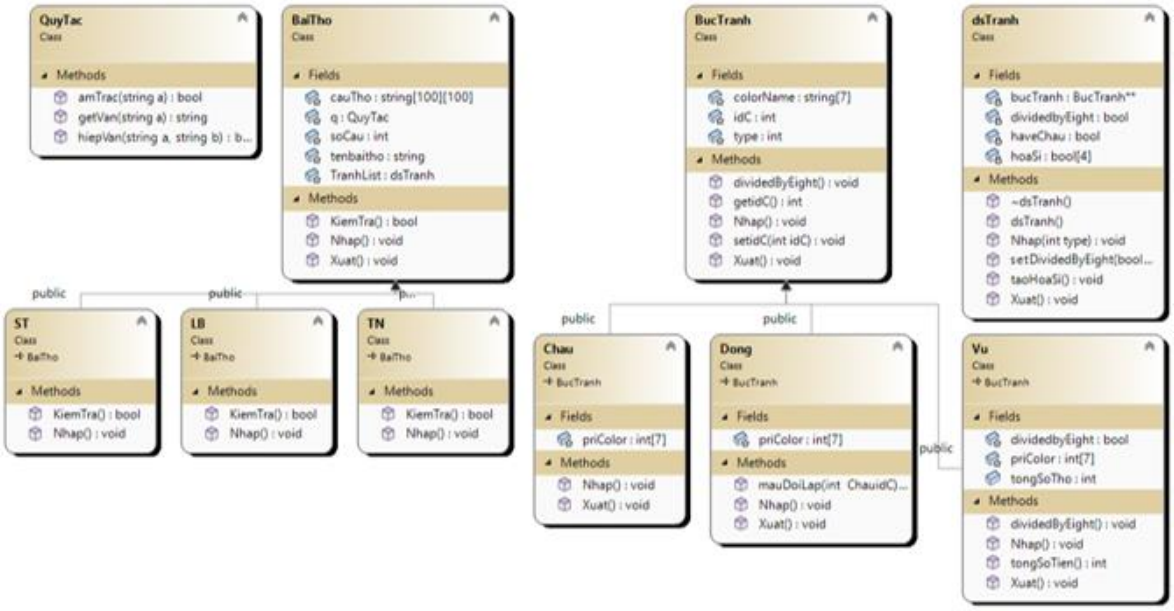
```
        return random(10, 15) * 6 * soTang;
    else return random(6, 10) * 6 * soTang;
}

CCBcon::CCBcon() {
    soHuu = "Bcons";
}
double CCBcon::tienPhong() {
    if (dienTich > 600)
        return random(8, 12) * 6 * soTang;
    else return random(5, 8) * 6 * soTang;
}

void ChungCu_import(ChungCu** x, int a, int b) {
    ChungCu* temp;
    cin.ignore();
    for (int i = 0; i < a; i++) {
        cout << "Thong tin chung cu VinHome thu " << i + 1 << ": " << endl;
        temp = new CCVinHome;
        temp->importInfo();
        x[i] = temp;
    }
    system("cls");
    for (int i = a; i < a + b; i++) {
        cout << "Thong tin chung cu Bcon thu " << i - a + 1 << ": " << endl;
        temp = new CCBcon;
        temp->importInfo();
        x[i] = temp;
    }
    system("cls");
}
void ChungCu_listExport(ChungCu** x, int size) {
    for (int i = 0; i < size; i++)
        cout << x[i]->guiThongBao() << endl;
}
void ChungCu_tienPhong(ChungCu** x, int size) {
    double total = 0;
    for (int i = 0; i < size; i++)
        total += x[i]->tienPhong();
    cout << "\nTong so tien thu ve moi thang cua ban quan ly chung cu lang dai hoc la " << total
<< "000 (VND)" << endl;
}
```

Bài 4.

Sơ đồ lớp:



Code:

```
#include <iostream>
#include <string>
using namespace std;

string getVan(string a) {
  int l = a.length();
  int i = 0;
  if (a[0] == 'q') i = 2;
  for (i; i < l; i++) {
    if (a[i] == 'a' || a[i] == 'u' || a[i] == 'i' || a[i] == 'e' || a[i] == 'o' || a[i] ==
'y') break;
  }
  string van = "";
  for (int j = i; j < l; j++) van.push_back(a[j]);
  return van;
}

bool amTrac(string a) {
  int l = a.length() - 1;
  return (a[l] == 'p' || a[l] == 'c' || a[l] == 't');
}

bool hiepVan(string a, string b) {
  string vanA = getVan(a);
  string vanB = getVan(b);
  int la = vanA.length() - 1;
  int lb = vanB.length() - 1;
  if (vanA[la] == 'g') vanA.pop_back();
  if (vanB[lb] == 'g') vanB.pop_back();
  if (vanA[la] == 't') vanA[la] = 'c';
  if (vanB[lb] == 't') vanB[lb] = 'c';
```

```
  if (vanA == "y") vanA = "i";
  if (vanB == "y") vanB = "i";
  if (vanA == vanB) return 1;
  return 0;
}

class BucTranh {

  protected: int type;
  string colorName[7] = {
    "",
    "Cam",
    "Lam",
    "Vang",
    "Luc",
    "Tim",
    "Do"
  };
  int idC = -1;

  public: virtual void Nhap() {
    cin >> idC;
  }
  virtual void Xuat() {}
  int getidC() {
    return idC;
  }
  void setidC(int idC) {
    this -> idC = idC;
  }
  virtual void dividedByEight() {}
};

class Chau: public BucTranh {

  protected: static int priColor[7];
  public: void Nhap() {
    type = 1;
    cout << "Mau chu dao cua Chau (1 - Cam, 2 - Lam, 3 - Vang, 4 - Luc, 5 - Tim, 6 - Do): ";
    BucTranh::Nhap();
    priColor[idC]++;

    if (idC == 2) priColor[2]++;

  }
  void Xuat() {

    cout << "Cac tranh cua Chau:\n";

    bool coTranh = 0;
    for (int i = 1; i < 7; i++) {
      if (priColor[i] > 0) {
        cout << colorName[i] << ": " << priColor[i] << " tranh\n";
        coTranh = 1;
```

```
      }
    }
    if (coTranh == 0) cout << "Khong ve tranh nao\n";

  }

};

class Dong: public BucTranh {

  protected: static int priColor[7];

  public: void Nhap() {
    type = 2;
    if (idC != -1) {
      mauDoiLap(idC);
    } else priColor[5]++;

  }
  void mauDoiLap(int ChaucrnColor) {
    switch (ChaucrnColor) {
    case 1:
      priColor[2]++;
      break;

    case 2:
      priColor[1]++;
      break;

    case 3:
      priColor[5]++;
      break;

    case 4:
      priColor[6]++;
      break;

    case 5:
      priColor[3]++;
      break;

    case 6:
      priColor[4]++;
      break;

    }
  }
  void Xuat() {

    cout << "Cac tranh cua Dong:\n";

    bool coTranh = 0;
    for (int i = 1; i < 7; i++) {
      if (priColor[i] > 0) {
```

```
        cout << colorName[i] << ": " << priColor[i] << " tranh\n";
        coTranh = 1;
      }
    }
    if (coTranh == 0) cout << "Khong ve tranh nao\n";
  }

};

class Vu: public BucTranh {

  protected: static int priColor[7];bool dividedbyEight = 0;

  public: static int tongSoTho;
  void Nhap() {

    type = 3;
    tongSoTho++;
    if (tongSoTho <= 2) {
      cout << "Mau chu dao cua Vu (1 - Cam, 2 - Lam, 3 - Vang, 4 - Luc, 5 - Tim, 6 - Do): ";
      BucTranh::Nhap();
      priColor[idC]++;
      if (dividedbyEight) priColor[idC]++;
    } else {
      cout << "Anh Vu da ve cho 2 bai tho\n";
    }
  }

  void dividedByEight() {
    dividedbyEight = 1;
  }
  int tongSoTien() {

    int tongSoTien = 0;
    for (int i = 1; i < 7; i++) {
      tongSoTien += 800 * priColor[i];
      if (i == 3) tongSoTien += 200 * priColor[i];
    }

    return tongSoTien;

  }

  void Xuat() {

    cout << "Cac tranh cua Vu:\n";

    bool coTranh = 0;
    for (int i = 1; i < 7; i++) {
      if (priColor[i] > 0) {
        cout << colorName[i] << ": " << priColor[i] << " tranh\n";
        coTranh = 1;
      }
    }
```

```
    if (coTranh == 0) cout << "Khong ve tranh nao\n";

    cout << "Tong chi phi: " << tongSoTien();

  }

};

int Vu::tongSoTho = 0;

int Chau::priColor[7] = {};
int Dong::priColor[7] = {};
int Vu::priColor[7] = {};

class dsTranh {
  private: bool hoaSi[4];BucTranh ** bucTranh = new BucTranh * [3];
  bool haveChau;bool dividedbyEight;
  public: dsTranh() {
      bucTranh[0] = new Chau;
      bucTranh[1] = new Dong;
      bucTranh[2] = new Vu;
    }
    ~dsTranh() {
      delete[] bucTranh;
    }
  void Nhap(int type = 0) {
    int temp;
    do {
      cin >> temp;
      hoaSi[temp] = 1;
    } while (cin.get() != '\n');
    if (type == 2) {
      hoaSi[2] = 0;
    }
    if (type == 3) {
      hoaSi[1] = 0;
    }
    taoHoaSi();
  }

  void setDividedByEight(bool dividedbyEight) {
    this -> dividedbyEight = dividedbyEight;
  }
  void taoHoaSi() {

    for (int i = 1; i < 4; i++) {
      if (hoaSi[i] == 1) {
        switch (i) {
        case 1:
          haveChau = 1;
          break;
        case 2:
          if (haveChau == 1) bucTranh[1] -> setidC(bucTranh[0] -> getidC());
          break;
```

```
        case 3:
          if (dividedbyEight) bucTranh[2] -> dividedByEight();
        }
        bucTranh[i - 1] -> Nhap();
      }
    }
  }

  void Xuat() {
    for (int i = 0; i < 3; i++) bucTranh[i] -> Xuat();
  }
};

class BaiTho {

  protected: string tenbaitho;
  string cauTho[100][100];
  int soCau = 0;
  dsTranh TranhList;

  public:

    virtual void Nhap() {
      cout << "- Ten bai tho: ";
      getline(cin, tenbaitho);
      cout << "- Nhap bai tho: \n";
      string temp;
      int chu = 1;
      do {
        cin >> temp;
        if (temp == "/") break;
        if (temp != " ") cauTho[soCau + 1][chu++] = temp;
        if (cin.get() == '\n') {
          soCau++;
          chu = 1;
        }
      } while (true);

      TranhList.setDividedByEight(soCau % 8 == 0);
    }

  virtual bool KiemTra() = 0;

  void Xuat() {

    if (KiemTra()) {
      cout << "==\n";
      cout << "-Tieu de: " << tenbaitho << "\n";
      for (int i = 1; i <= soCau; i++) {
        int chu = 1;
        while (cauTho[i][chu] != "") {
          cout << cauTho[i][chu++] << " ";
        }
        cout << "\n";
```

```
      }
    }
  }

};

class LB: public BaiTho {

  public: void Nhap() {
    BaiTho::Nhap();
    if (!KiemTra()) return;
    cout << "- Cac hoa si muon nho (1 - Chau, 2 - Dong, 3 - Vu): ";
    TranhList.Nhap(1);
  }

  bool KiemTra() {

    if (soCau % 2 != 0) return 0;
    for (int i = 1; i <= soCau; i++) {
      if (i % 2 == 1 && (cauTho[i][7] != "")) return 0;
      if (i % 2 == 0 && (cauTho[i][9] != "")) return 0;
      if (i % 2 == 1 && !hiepVan(cauTho[i][6], cauTho[i + 1][6])) return 0;
      if (i % 2 == 0 && i != soCau && !hiepVan(cauTho[i][8], cauTho[i + 1][6])) return 0;
    }
    return 1;
  }

};

class ST: public BaiTho {

  public: void Nhap() {
    BaiTho::Nhap();
    if (!KiemTra()) return;
    cout << "- Cac hoa si muon nho (1 - Chau, 3 - Vu): ";
    TranhList.Nhap(2);

  }

  bool KiemTra() {
    if (soCau % 4 != 0) return 0;
    int i = 1;
    if (!amTrac(cauTho[1][7])) return 0;
    if (amTrac(cauTho[2][7])) return 0;
    if (amTrac(cauTho[4][8])) return 0;
    while (i <= soCau - 3) {
      if ((i % 4 == 1 || i % 4 == 2) && (cauTho[i][8] != "")) return 0;
      if (i % 4 == 3 && (cauTho[i][7] != "")) return 0;
      if (i % 4 == 0 && (cauTho[i][9] != "")) return 0;
      if (!hiepVan(cauTho[i][7], cauTho[i + 1][5])) return 0;
      if (!hiepVan(cauTho[i + 1][7], cauTho[i + 2][6])) return 0;
      if (!hiepVan(cauTho[i + 2][6], cauTho[i + 3][6])) return 0;
      if ((i + 4 < soCau) && (!hiepVan(cauTho[i + 3][8], cauTho[i + 4][5]))) return 0;
      i += 4;
```

```
    }
    return 1;

  }

};

class TN: public BaiTho {

  public: void Nhap() {
    BaiTho::Nhap();
    if (!KiemTra()) return;
    cout << "- Cac hoa si muon nho (2 - Dong, 3 - Vu): ";
    TranhList.Nhap(3);
  }

  bool KiemTra() {
    if (soCau != 8) return 0;
    for (int i = 1; i <= 8; i++) {
      if (cauTho[i][8] != "") return 0;
    }

    if (amTrac(cauTho[1][7])) return 0;
    if (!hiepVan(cauTho[1][7], cauTho[2][7])) return 0;
    if (!hiepVan(cauTho[2][7], cauTho[4][7])) return 0;
    if (!hiepVan(cauTho[4][7], cauTho[6][7])) return 0;
    if (!hiepVan(cauTho[6][7], cauTho[8][7])) return 0;
    return 1;
  }

};

int main() {

  cout << "- Nhap so luong bai tho: ";
  int n;
  cin >> n;
  BaiTho ** tho = new BaiTho * [n];
  for (int i = 0; i < n; i++) {

    cout << "== The loai bai tho thu " << i + 1 << "(LB - Luc bat, ST - Song that luc bat, TN
- That ngon bat cu Duong luat): ";

    string type;
    cin >> type;
    cin.ignore();

    if (type == "LB") tho[i] = new LB;
    if (type == "ST") tho[i] = new ST;
    if (type == "TN") tho[i] = new TN;

    tho[i] -> Nhap();

  }
```

```
  for (int i = 0; i < n; i++) {
    tho[i] -> Xuat();
  }

  cout << "--\n";
  dsTranh ds;
  ds.Xuat();
  return 0;

}
```